Math Thermo
class # 28
5/07/2026

The Cahn – Larche – Fourier Model

Assume that density of a binary ($r=2$) mixture is constant (approximately):

$$\rho \equiv \rho_0 > 0 .$$

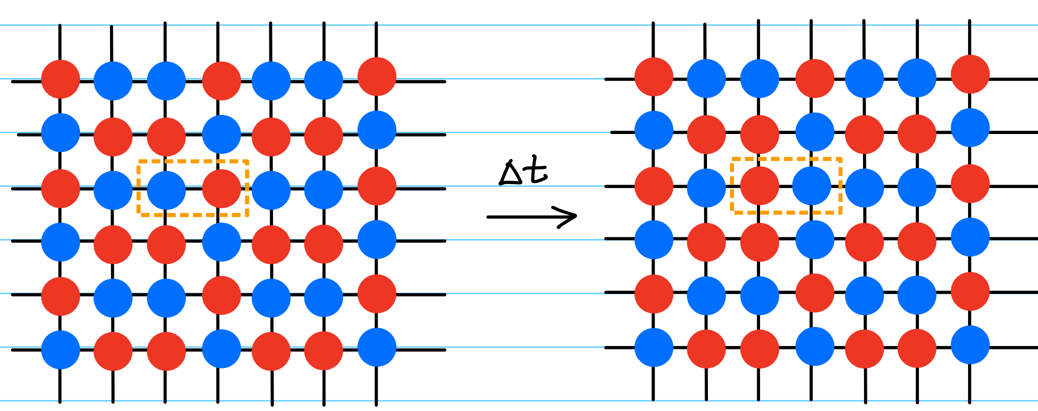
We will use a technique similar to that used in the last example. Assuming that the flow is not reactive and we are operating in the diffusion dominated regime, we have the following evolution equations

$$\frac{\partial \phi}{\partial t} = -\frac{1}{\rho_0} \nabla \cdot \vec{J},$$

and

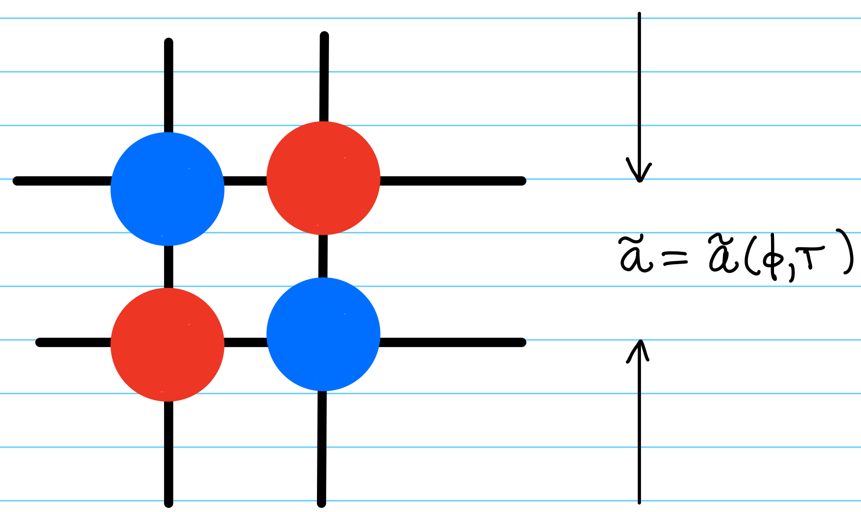
$$\frac{\partial \hat{u}}{\partial t} = -\nabla \cdot \vec{J}_u^D .$$

We assume that the atoms in the binary mixture diffuse on a lattice



Now, in slight contradiction to our earlier assumption, we will assume that the lattice spacing is a function of ϕ and T:

$$\tilde{a} = \tilde{a}(\phi, T).$$



Near to some reference state, (ϕ_0, T_0), we can approximate $\tilde{a}$ by a linear function. Set

$$\tilde{\alpha}_0 = \tilde{a}(\phi_0, T_0).$$

A good approximation of $\tilde{a}$ is

$$a(\phi, T) = \tilde{\alpha}_0 + \tilde{\alpha}_1(\phi - \phi_0) + \tilde{\alpha}_2(T - T_0),$$

where $\tilde{\alpha}_1 \in \mathbb{R}$ is a constant and $\tilde{\alpha}_2 > 0$ is a positive constant.

Define

$$e(\phi, T) := \frac{a(\phi, T) - \tilde{\alpha}_0}{\tilde{\alpha}_0}.$$

This is called the (unitless) *misfit strain*. Clearly,

$$e(\phi, T) = \frac{\tilde{\alpha}_1}{\tilde{\alpha}_0}(\phi - \phi_0) + \frac{\tilde{\alpha}_2}{\tilde{\alpha}_0}(T - T_0)$$

$$= \alpha_1(\phi - \phi_0) + \alpha_2(T - T_0),$$

where

$$\alpha_i = \frac{\tilde{\alpha}_i}{\tilde{\alpha}_0}, \quad i = 1, 2.$$

We use the model

$$\hat{f}(T, \phi, \nabla\phi, \nabla\vec{u}) = \tilde{f}(T) + \lambda T \delta(\phi, \nabla\phi) + T G(\phi, T, \nabla\vec{u}),$$

where

$$\delta = \frac{1}{\varepsilon} W(\phi) + \frac{\varepsilon}{2}|\nabla\phi|^2$$

and $\lambda > 0$ is a constant. G is called the elastic energy density and is defined as

(28.1) $$G(\phi, T, \nabla\vec{u}) = \frac{1}{2}\sigma_{ij}(\epsilon_{ij} - e(\phi, T)\delta_{ij}),$$

where $\epsilon = [\epsilon_{ij}]_{i,j=1}^{3}$, $\epsilon_{ij} := \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$,

is the *elastic strain tensor*; $\vec{u} = (u_i)_{i=1}^{3}$ is is *displacement vector*; and σ_{ij} is the *stress tensor*. The stress tensor, in linear elasticity theory, is related to the strain tensor via

(28.2) $$\sigma_{i,j} = C_{i,j,k,l}\left(\epsilon_{k,l} - e(\phi,T)\,\delta_{k,l}\right),$$

which is called <u>Hooke's law</u>.

In equations (28.1) and (28.2) we use Einstein's summation convention.

The stress tensor is symmetric, and this requirement places certain restrictions on the elastic stiffness tensor $C_{i,j,k,l}$, which, generally, has 81 entries.

<u>Proposition</u> (28.1): Since $\epsilon_{i,j}$ is symmetric,

(28.3) $$C_{i,j,k,l} = C_{i,j,l,k}, \qquad \forall\, i,j,k,l \in \{1,2,3\}.$$

Since $\sigma_{i,j}$ is symmetric

(28.4) $$C_{i,j,k,l} = C_{j,i,k,l}, \qquad \forall\, i,j,k,l \in \{1,2,3\}.$$

<u>Proof</u>: Exercise. ///

We further assume that

(28.5) $$C_{i,j,k,l} = C_{k,l,i,j}, \qquad \forall\, i,j,k,l \in \{1,2,3\}.$$

Symmetries (28.3) and (28.4) are called <u>left and right minor symmetries</u>, respectively; (28.5) is called <u>major symmetry</u>. Finally assume <u>positive definiteness</u>,

(28.6) $$\epsilon_{i,j}\, C_{i,j,k,l}\, \epsilon_{k,l} > 0,$$

for all symmetric two-tensors $\epsilon \neq 0$.

Thus, we see that

$$G(\phi, T, \nabla\vec{u}) = \frac{1}{2}(\epsilon_{i,j} - e\delta_{i,j})\, C_{i,j,k,l}(\epsilon_{k,l} - e\delta_{k,l})$$

$$\geq 0,$$

with equality iff

$$\epsilon_{i,j} = e\,\delta_{i,j}, \qquad \forall\, i,j \in \{1,2,3\}.$$

G measures the elastic energy stored in the lattice due to misfit-strain-induced deformations.

Now, we need to compute the internal energy density, $\hat{u}$, and the entropy density, $\hat{s}$. We will use the following.

<u>Theorem</u> (28.1): Suppose that $\hat{f}(T, \phi, \nabla\phi, \nabla\vec{u})$ is given,

(28.7) $$\hat{s} = -\frac{\partial \hat{f}}{\partial T}$$

and

(28.8) $$\hat{u} = \hat{f} + T\hat{s}.$$

Then thermal compatibility holds, i.e.,

(28.9) $$\frac{\partial \hat{s}}{\partial T} = \frac{1}{T}\frac{\partial \hat{u}}{\partial T}.$$

<u>Proof</u>: This is a simple calculation.

Observe that
$$\frac{\partial \hat{u}}{\partial T} = \frac{\partial \hat{f}}{\partial T} + \hat{s} + T\frac{\partial \hat{s}}{\partial T}$$
$$= \frac{\partial \hat{f}}{\partial T} - \frac{\partial \hat{f}}{\partial T} + T\frac{\partial \hat{s}}{\partial T}$$
$$= T\frac{\partial \hat{s}}{\partial T}. \quad ///$$

Since we know the form of $\hat{f}$ precisely, we can easily establish $\hat{s}$ and $\hat{u}$, which are needed to construct our model, using the method suggested above.

To complete $\hat{f}$, let us assume that
$$\tilde{f}(T) = -\rho_0 C T \log\left(\frac{T}{T_0}\right).$$

Then,
$$\hat{s} = \rho_0 C \log\left(\frac{T}{T_0}\right) + \rho_0 C - \lambda\,\delta(\phi, \nabla\phi) - G - T\frac{\partial G}{\partial T}.$$

Now, let us find the derivative of the elastic energy density:
$$2\frac{\partial G}{\partial T} = \frac{d}{dT}\left\{C_{i,j,k,l}(\epsilon_{k,l} - e\,\delta_{k,l})(\epsilon_{i,j} - e\,\delta_{i,j})\right\}$$
$$= \frac{d}{dT}\left\{(\epsilon_{i,j} - e\,\delta_{i,j})C_{i,j,k,l}(\epsilon_{k,l} - e\,\delta_{k,l})\right\}$$
$$= -\alpha_2 C_{i,i,k,l}(\epsilon_{k,l} - e\,\delta_{k,l})$$
$$- \alpha_2(\epsilon_{i,j} - e\,\delta_{i,j})C_{i,j,k,k}$$
$$= -2\alpha_2 C_{i,i\,k,l}(\epsilon_{k,l} - e\,\delta_{k,l}).$$

Thus,
$$\frac{\partial G}{\partial t} = -\alpha_2\,\sigma_{i,i}.$$

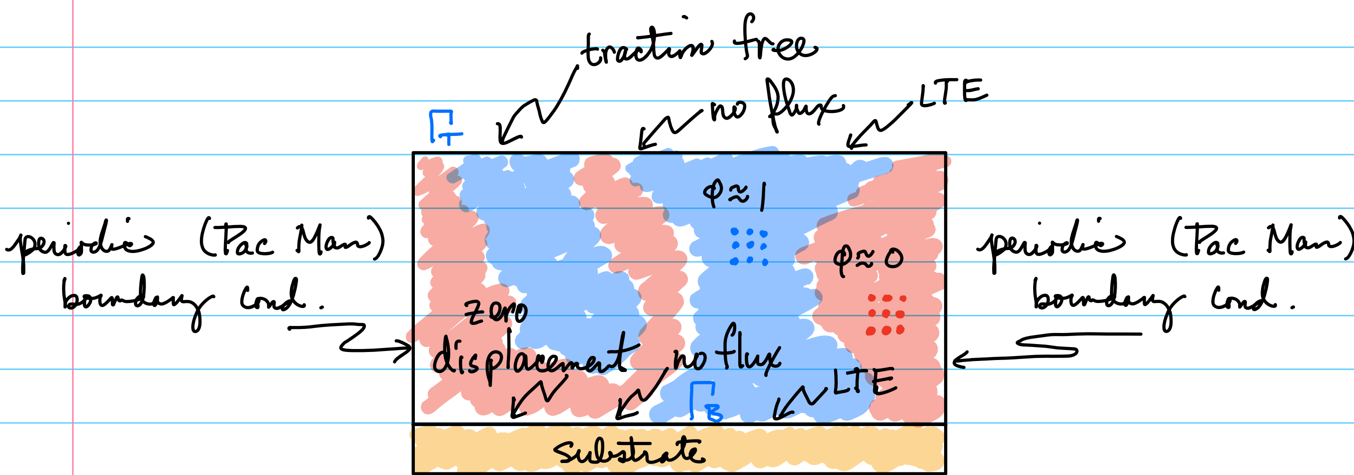
Finally,

$$\hat{u} = -\cancel{\rho_0 C T \log}\left(\frac{T}{T_0}\right) + \lambda T \cancel{\delta(\phi, \nabla\phi)}$$
$$+ T G(\phi, T, \nabla\vec{u}) + T\Big\{ \cancel{\rho_0 C \log}\left(\frac{T}{T_0}\right)$$
$$+ \rho_0 C - \lambda \cancel{\delta(\phi, \nabla\phi)} - G - T\frac{\partial G}{\partial T}\Big\}$$

so that

$$\hat{u} = \rho_0 C T - T^2 \frac{\partial G}{\partial T}.$$

For the geometry and boundary conditions, we will assume what is usual for a thin film system:



The boundary conditions are

$$\sigma_{ij} n_j = 0 \quad \text{on } \Gamma_T, \quad \text{(traction free)}$$

$$\frac{\partial \phi}{\partial n} = 0 \quad \text{on } \Gamma_T \,\&\, \Gamma_B, \quad \text{(LTE)}$$

$$\vec{J} \cdot \hat{n} = 0 \quad \text{on } \Gamma_T \,\&\, \Gamma_B, \quad \text{(no mass flux)}$$

$$\vec{J}_U^{\,D} \cdot \hat{n} = 0 \quad \text{on } \Gamma_T \,\&\, \Gamma_B, \quad \text{(no heat flux)}$$

$$u_i = 0 \quad \text{on } \Gamma_B. \quad \text{(zero displacement)}$$

We use the integral technique, since we only need to identify σ_S.

$$S = \int_\Omega \hat{s}(T, \phi, \nabla\phi, \nabla\vec{u})\, d\vec{x}$$

Let us represent the time derivative by an over dot:

$$\dot{S} = \frac{dS}{dt} = d_t S .$$

Recall

$$\hat{s} = \rho_0 C \log\left(\frac{T}{T_0}\right) + \rho_0 C - \lambda \delta(\phi, \nabla\phi) - G - T\frac{\partial G}{\partial T}$$

and

$$\hat{u} = \rho_0 C T - T^2 \frac{\partial G}{\partial T}.$$

$$\frac{\partial \hat{s}}{\partial T} = \rho_0 C \frac{1}{T} - \frac{\partial G}{\partial T} - \frac{\partial G}{\partial T} - T\frac{\partial^2 G}{\partial T^2};$$

$$\frac{1}{T}\frac{\partial \hat{u}}{\partial T} = \frac{1}{T}\left\{\rho_0 C - 2T\frac{\partial G}{\partial T} - T^2 \frac{\partial^2 G}{\partial T^2}\right.$$

$$= \rho_0 C \frac{1}{T} - 2\frac{\partial G}{\partial T} - T\frac{\partial^2 G}{\partial T^2}.$$

Thermal compatibility holds, as desired.

We have, by the chain rule,

$$\dot{S} = \int_\Omega \left\{ \frac{\partial \hat{s}}{\partial T}\dot{T} + \frac{\partial \hat{s}}{\partial \phi}\dot{\phi} + \frac{\partial \hat{s}}{\partial p_i}\dot{p}_i + \frac{\partial \hat{s}}{\partial w_{i,j}}\dot{w}_{i,j} \right\} d\vec{x}$$

where

$$p_i = \frac{\partial \phi}{\partial x_i},$$

$$w_{i,j} = \frac{\partial u_i}{\partial x_j},$$

and the Einstein summation convention is used. Using thermal compatibility,

$$\dot{S} = \int_{\Omega} \left\{ \frac{1}{T} \frac{\partial \hat{u}}{\partial T} \dot{T} + \frac{\partial \hat{s}}{\partial \phi} \dot{\phi} + \frac{\partial \hat{s}}{\partial p_i} \dot{p}_i + \frac{\partial \hat{s}}{\partial W_{ij}} \dot{W}_{ij} \right\} d\vec{x}.$$

Now, we need the form for

$$\frac{\partial \hat{u}}{\partial t} = -\nabla \cdot \vec{J}_U^D.$$

Generically,

$$\frac{\partial \hat{u}}{\partial t} = \frac{\partial \hat{u}}{\partial T} \dot{T} + \frac{\partial \hat{u}}{\partial \phi} \dot{\phi} + \frac{\partial \hat{u}}{\partial p_i} \dot{p}_i + \frac{\partial \hat{u}}{\partial W_{ij}} \dot{W}_{ij}.$$

But note that

$$\frac{\partial \hat{u}}{\partial p_i} = 0, \quad i = 1, 2, 3,$$

since $\hat{u}$ does not depend, in this special case, upon $\nabla \phi$. Thus, the heat equation becomes

$$\frac{\partial \hat{u}}{\partial T} \dot{T} = -\nabla \cdot \vec{J}_U^D - \frac{\partial \hat{u}}{\partial \phi} \dot{\phi} - \frac{\partial \hat{u}}{\partial p_i} \dot{p}_i - \frac{\partial \hat{u}}{\partial W_{ij}} \dot{W}_{ij}.$$

We find that

(28.10) $$\dot{S} = \int_{\Omega} \frac{1}{T} \left\{ -\nabla \cdot \vec{J}_U^D - \frac{\partial \hat{f}}{\partial \phi} \dot{\phi} - \frac{\partial \hat{f}}{\partial p_i} \dot{p}_i - \frac{\partial \hat{f}}{\partial W_{ij}} \dot{W}_{ij} \right\} d\vec{x},$$

where we have used

$$T\hat{s} - \hat{u} = -\hat{f}.$$

The last equation, Equation (28.10), is actually quite general, using only the structure implied by Theorem (28.1)!

Note that, using integration-by-parts,

$$-\int_{\Omega} \frac{1}{T} \frac{\partial \hat{f}}{\partial p_i} \dot{p}_i \, d\vec{x} = -\int_{\Omega} \frac{1}{T} \frac{\partial \hat{f}}{\partial p_i} \frac{\partial}{\partial x_i}\left(\frac{\partial \phi}{\partial t}\right) d\vec{x}$$

$$= \int_{\Omega} \frac{\partial}{\partial x_i}\left(\frac{1}{T}\frac{\partial \hat{f}}{\partial p_i}\right) \frac{\partial \phi}{\partial t} \, d\vec{x}$$

$$-\int_{\partial\Omega} \frac{1}{T} \frac{\partial \hat{f}}{\partial p_i} n_i \frac{\partial \phi}{\partial t} \, dS .$$

Likewise,

$$-\int_{\Omega} \frac{1}{T} \frac{\partial \hat{f}}{\partial w_{i,j}} \dot{w}_{i,j} \, d\vec{x} = -\int_{\Omega} \frac{1}{T} \frac{\partial \hat{f}}{\partial w_{i,j}} \frac{\partial}{\partial x_j}\left(\frac{\partial u_i}{\partial t}\right) d\vec{x}$$

$$= \int_{\Omega} \frac{\partial}{\partial x_j}\left(\frac{1}{T}\frac{\partial \hat{f}}{\partial w_{i,j}}\right) \frac{\partial u_i}{\partial t} \, d\vec{x}$$

$$-\int_{\partial\Omega} \frac{1}{T} \frac{\partial \hat{f}}{\partial w_{i,j}} n_j \frac{\partial u_i}{\partial t} \, dS .$$

Thus,

$$\dot{S} = \int_{\Omega} \Big\{ -\frac{\nabla \cdot \vec{J}_U{}^{D}}{T} - \left[\frac{1}{T}\frac{\partial \hat{f}}{\partial \phi} - \frac{\partial}{\partial x_i}\left(\frac{1}{T}\frac{\partial \hat{f}}{\partial p_i} \right) \right] \dot{\phi}$$

$$+ \frac{\partial}{\partial x_j}\left(\frac{1}{T}\frac{\partial \hat{f}}{\partial w_{i,j}} \right) \dot{u}_i \Big\} \, d\vec{x}$$

$$- \int_{\partial\Omega} \frac{1}{T}\frac{\partial \hat{f}}{\partial p_i} n_i \, \dot{\phi} \, ds$$

$$- \int_{\Omega} \frac{1}{T}\frac{\partial \hat{f}}{\partial w_{i,j}} n_j \frac{\partial u_i}{\partial t} \, d\vec{x} .$$

It is not hard to show that, after writing

$$T \frac{\lambda \varepsilon}{2} |\nabla \phi|^2 = T \frac{\lambda \varepsilon}{2} p_i \, p_i ,$$

$$\frac{1}{T}\frac{\partial \hat{f}}{\partial p_i} = \frac{T \lambda \varepsilon}{T}\frac{\partial \phi}{\partial x_i} \Rightarrow \frac{\partial}{\partial x_i}\left(\frac{1}{T}\frac{\partial \hat{f}}{\partial p_i} \right) = \varepsilon \lambda \Delta \phi$$

The term $\frac{\partial \hat{f}}{\partial w_{i,j}}$ is a bit harder to compute.
We have

$$\frac{\partial \hat{f}}{\partial w_{i,j}} = T \frac{\partial G}{\partial w_{i,j}}$$

$$= \frac{T}{2} \frac{\partial}{\partial w_{i,j}} \left[(\epsilon_{k,l} - e \, \delta_{k,l}) \, C_{k,l,m,n} \, (\epsilon_{m,n} - e \, \delta_{m,n}) \right]$$

$$= \frac{T}{2} \frac{\partial \epsilon_{k,l}}{\partial w_{i,j}} C_{k,l,m,n} \left(\epsilon_{m,n} - e \, \delta_{m,n} \right)$$

$$+ \frac{T}{2} \left(\epsilon_{k,l} - e \, \delta_{k,l} \right) C_{k,l,m,n} \frac{\partial \epsilon_{m,n}}{\partial w_{i,j}}$$

$$= T \frac{\partial \epsilon_{k,l}}{\partial W_{i,j}} \sigma_{k,l} .$$

Now,

$$\frac{\partial \epsilon_{k,l}}{\partial W_{i,j}} = \frac{1}{2} \left(\delta_{i,k} \delta_{j,l} + \delta_{i,l} \delta_{j,k} \right).$$

Thus,

$$\frac{\partial \hat{f}}{\partial W_{i,j}} = \frac{T}{2} \delta_{i,k} \delta_{j,l} \sigma_{k,l} + \frac{T}{2} \delta_{i,l} \delta_{j,k} \sigma_{k,l}$$

$$= \frac{T}{2} \sigma_{i,j} + \frac{T}{2} \sigma_{j,i}$$

$$= T \sigma_{i,j} .$$

So, to sum up

$$\dot{S} = \int_\Omega \left\{ \frac{-\nabla \cdot \vec{J}_U{}^D}{T} + \overbrace{\left[\frac{1}{T} \frac{\partial \hat{f}}{\partial \phi} - \varepsilon \lambda \Delta \phi \right]}^{=:\omega} \frac{\nabla \cdot \vec{J}}{\rho_0} \right.$$

$$\left. + \frac{\partial}{\partial x_j} \left(\frac{\cancel{T}}{\cancel{T}} \sigma_{i,j} \right) \dot{u}_i \right\} d\vec{x}$$

$$- \int_{\partial\Omega} \frac{\cancel{T}}{\cancel{T}} \lambda \varepsilon \frac{\partial \phi}{\partial x_i} n_i \dot{\phi} \, ds$$

$$- \int_\Omega \frac{\cancel{T}}{\cancel{T}} \sigma_{i,j} n_j \frac{\partial u_i}{\partial t} d\vec{x} .$$

Owing to our boundary conditions, the boundary integrals vanish and

$$\dot{S} = \int_{\Omega} \left\{ -\frac{\nabla \cdot \vec{J}_U^{\,D}}{T} + \omega \frac{\nabla \cdot \vec{J}}{\rho_0} + (\nabla \cdot \sigma) \cdot \dot{\vec{u}} \right\} d\vec{x}$$

We assume that the elasticity problem is quasi-static. That is, we assume that elastic equilibrium holds:

$$\nabla \cdot \sigma = \vec{0}$$

or

$$\frac{\partial}{\partial x_i} \sigma_{ij} = 0, \qquad j = 1, 2, 3.$$

Thus,

$$\dot{S} = \int_{\Omega} \left\{ \nabla\left(\frac{1}{T}\right) \cdot \vec{J}_U^{\,D} - \nabla \omega \cdot \vec{J} \right\} d\vec{x},$$

upon using IBP and the no-flux boundary conditions.

The simplest entropy-producing model is

$$\vec{J}_U^{\,D} = M_T T^2 \nabla\left(\frac{1}{T}\right), \qquad M_T > 0,$$

and

$$\vec{J} = -M_\phi \nabla \omega, \quad M_\phi > 0,$$

for which

$$\dot{S} = \int_{\Omega} \sigma_s \, d\vec{x}$$

with

$$\sigma_s = M_T T^2 \left|\nabla\left(\frac{1}{T}\right)\right|^2 + M_\phi |\nabla \omega|^2 \geq 0.$$